Distributed Muon at Scale

Sharding a matrix optimizer across thousands of GPUs

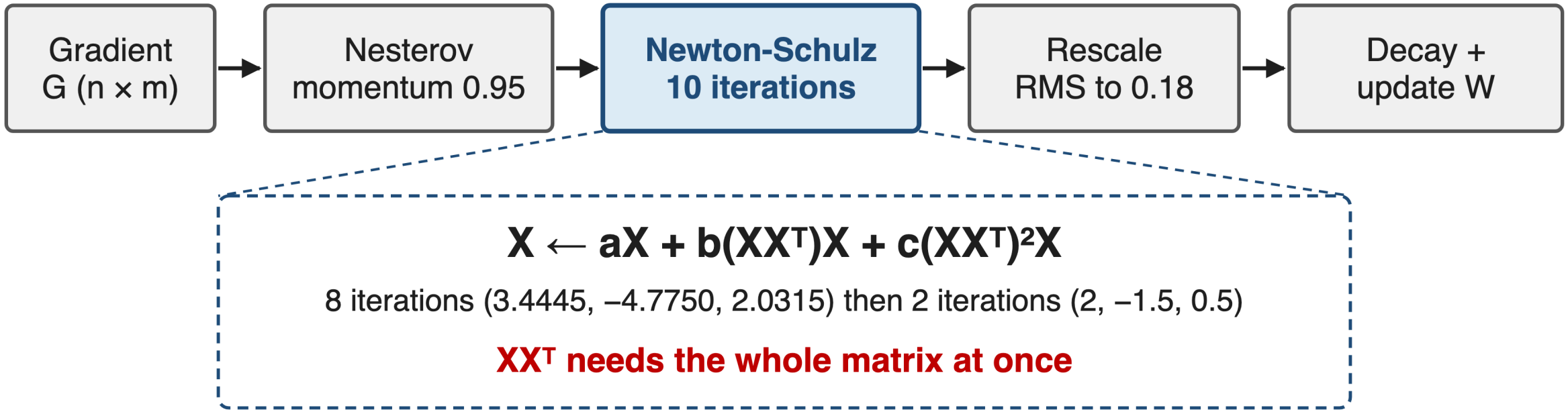
DeepSeek-AI · DeepSeek-V4 (also GLM-5, Moonlight)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Optimizer & Stability · arxiv.org/abs/2606.19348

Muon updates each weight matrix as one unit

per logically independent matrix W



Muon trains V4, K2 and GLM-5; Moonlight measured ~2× AdamW's efficiency

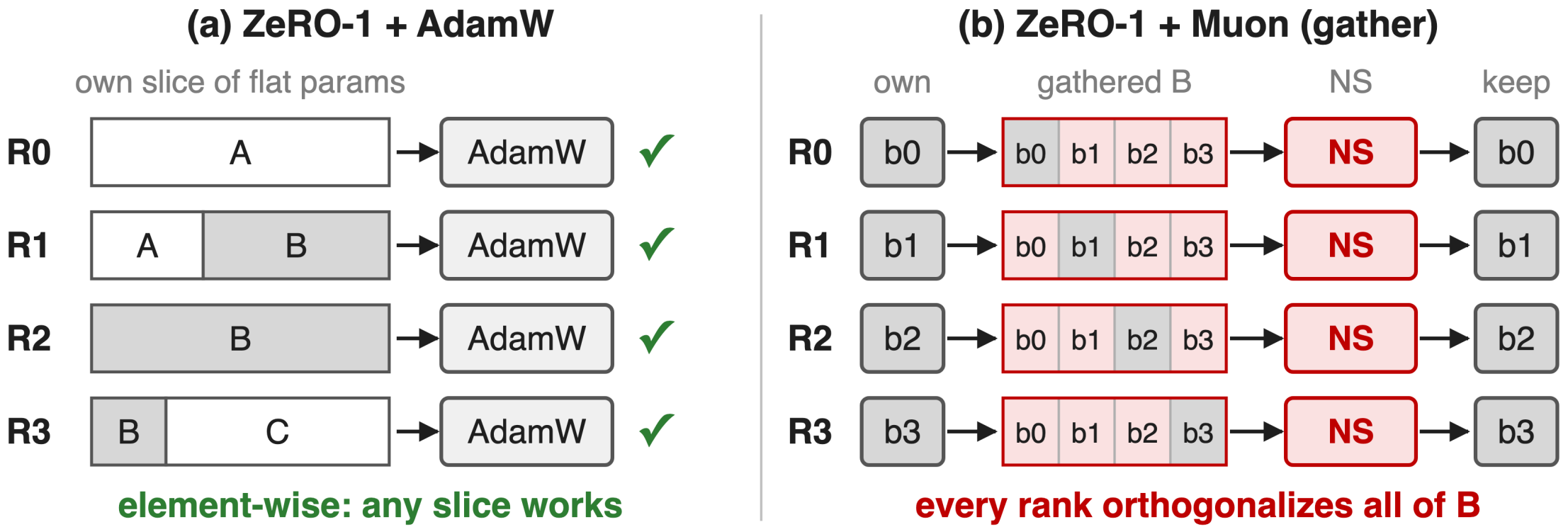
Gathering for Muon costs up to 1.25× traffic

1.25×

DP traffic of ZeRO-1 "Distributed Muon" vs. ZeRO-1 AdamW, and every rank still runs Newton-Schulz on the full matrix

Moonlight, arXiv 2502.16982, §2.3 (4 + 2 + 4 vs. 4 + 4 bytes per weight); GLM-5, arXiv 2602.15763, §2.4.1

ZeRO slices the matrices Muon needs whole



ZeRO shards elements; Muon's unit of work is a matrix

Insight: shard by matrix, not by element

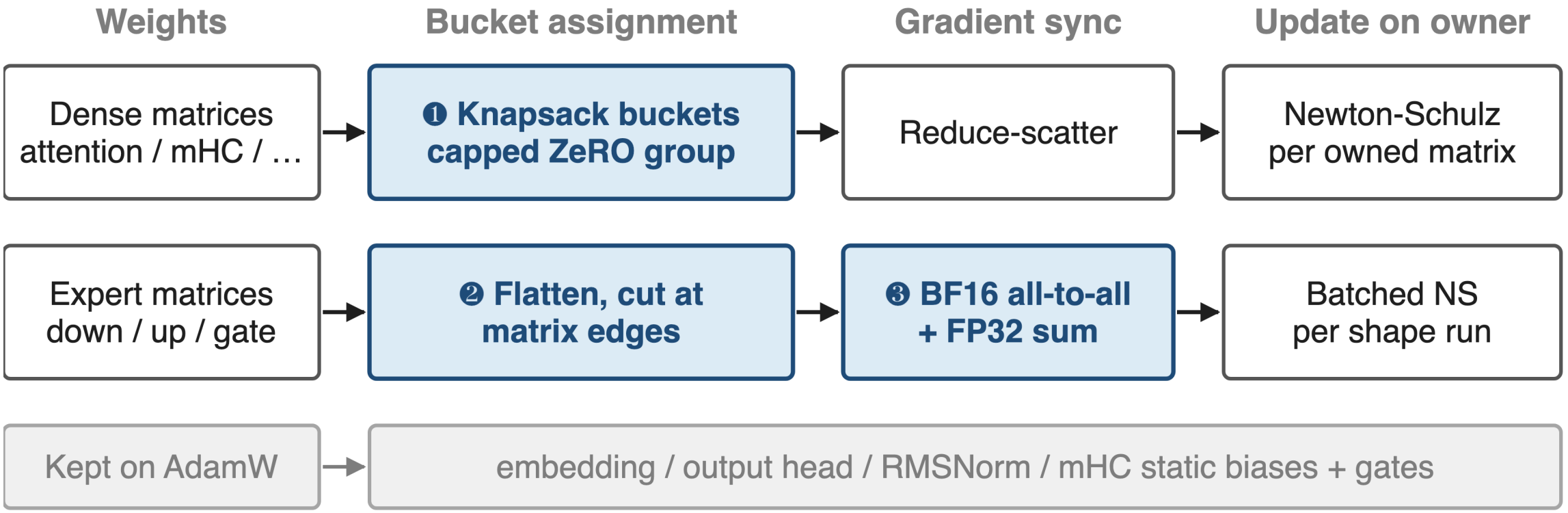
- V4-Pro: **70,455** expert matrices of **22M** weights each (derived)
- Far more matrices than ranks: whole ones balance by count

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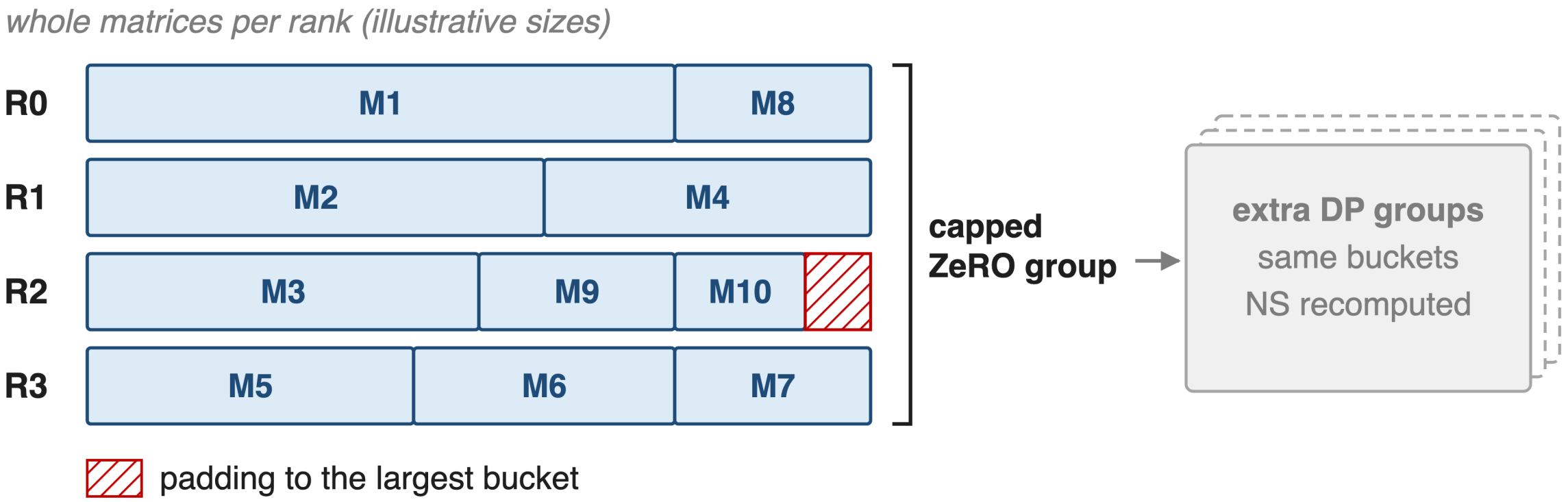
Give each rank whole matrices; balance by packing, not by cutting

V4 routes dense and expert weights differently



Every matrix lives whole on one owner, which runs its Newton-Schulz

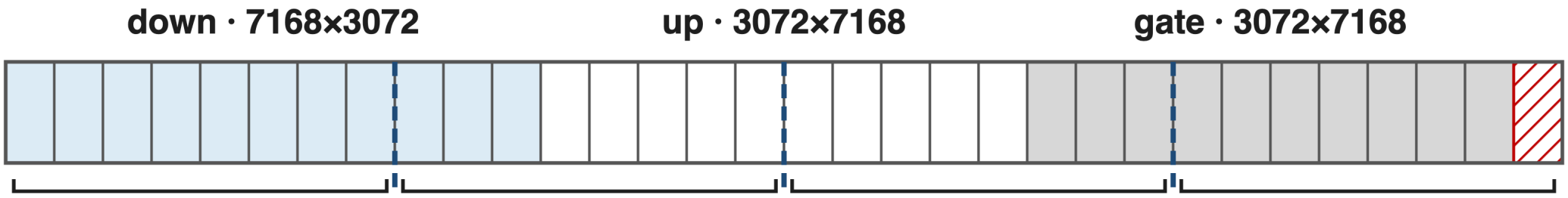
❶ Knapsack packs whole dense matrices per rank



At most 5 matrices per rank keeps padding under 10% of memory

❷ Expert matrices split only at matrix edges

expert matrices of all layers, flattened (illustrative count)



Same-shape neighbors merge into one batched Newton-Schulz call

❸ BF16 on the wire, FP32 in the sum

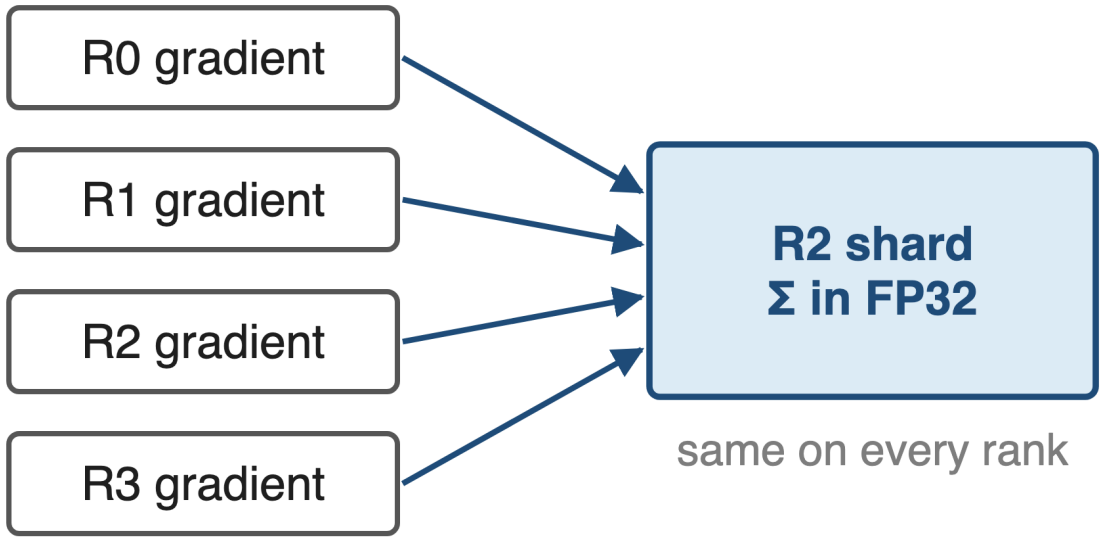
(a) Ring reduce-scatter in BF16



rounding error at every hop

(b) V4: all-to-all, then FP32 sum

BF16 with stochastic rounding

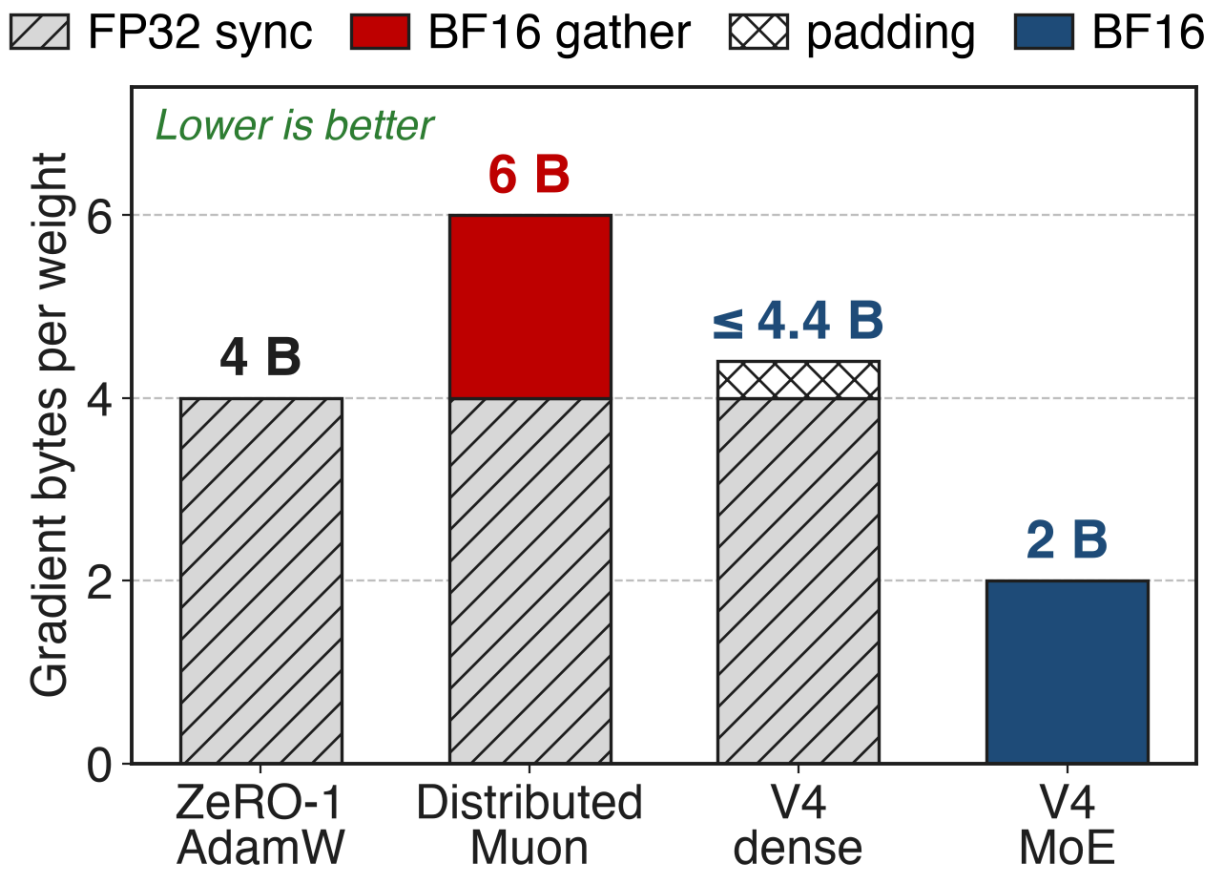


one rounding, FP32 accumulation

Newton-Schulz stays stable in BF16, so expert gradients travel in BF16

Gradient traffic falls from 6 to 2 B/weight

- No gather: owners receive whole matrices
- Expert sync in BF16: **half** the bytes
- Dense: padding under **10%**



Analytic: gradient path only, one ZeRO group; FP32 = 4 B, BF16 = 2 B

GLM-5 removes the redundant all-gather instead

- **Naive Muon**
- Every DP rank all-gathers full params
- Transient memory spikes, redundant traffic

- **GLM-5**
- All-gather only the shards each rank owns
- Overlap shard traffic with local compute

Same rule as V4: no rank materializes what it does not own

The cost: recompute, padding, BF16 trust

- **Recompute**: dense NS repeated in every extra DP group
- **Padding**: under 10% only with ≤ 5 matrices per rank
- **BF16 trust**: NS stability is observed, not proven
- AdamW still covers embedding, head, norms
- Tensor parallelism adds a BF16 gather

Summary

- ZeRO slices matrices: up to **1.25×** traffic, NS on every rank
- **Insight**: shard by matrix; **70,455** expert matrices balance by count
- Knapsack dense buckets (< **10%** padding); experts cut at matrix edges
- Expert gradients in BF16: **6** → **2** bytes per weight (analytic)

https://arxiv.org/abs/2606.19348